Minime CI Workflow Architecture (Vertical Pipeline)

Unified GitHub Actions Build System — `.github/workflows/build.yml` [Shared]
Legend: [Shared] = Dual-distro / shared across targets | [Specific] = Single-distro, board, or libc specific

1. Trigger / Entrypoint

- **Workflow File:** `.github/workflows/build.yml` [Shared]
- **Triggers:** Push to `main`, Pull Request, or `workflow_dispatch`

↓ *(Triggers parallel Tier 1 jobs)*

2. Parallel Tier 1 Builds (Bootloader & UIs)

2A. Bootloader Build

build-bootloader (AMD64)
- **Workflow Job:** `build-bootloader` [Shared]
- **Runner:** `ubuntu-latest`
- **Container:** `minime-glibc:latest` [Shared]
- **Script:** `scripts/build-bootloader.sh` [Shared]
- **Cache Key:** `bootloader-*` (hash of `uboot/**`)
- **Configs & Patches:**
  - `minime/uboot/config/uboot.config` [Shared]
  - `minime/uboot/config/ddr3.defconfig` [Specific] (H700)
  - `minime/uboot/out/rk3566/rkbin/` [Specific] (RK3566)
- **GH Artifact:** `bootloader-out`

2B. UI Build (musl)

build-ui (musl) (ARM64)
- **Workflow Job:** `build-ui (musl)` [Shared]
- **Runner:** `ubuntu-24.04-arm`
- **Container:** `minime-musl:latest` [Shared]
- **Script:** `scripts/build-ui.sh minui|allium musl` [Shared]
- **Cache Key:** `ui-musl-*` (hash of `ui/**`)
- **Submodules:**
  - `minime/ui/minui/` [Shared] (dual musl/glibc)
  - `minime/ui/allium/` [Shared] (dual musl/glibc)
- **GH Artifact:** `ui-musl`
  - `minui-musl-aarch64.tar.xz`
  - `allium-musl-aarch64.tar.xz`

2C. UI Build (glibc)

build-ui (glibc) (AMD64)
- **Workflow Job:** `build-ui (glibc)` [Shared]
- **Runner:** `ubuntu-latest`
- **Container:** `minime-glibc:latest` [Shared]
- **Script:** `scripts/build-ui.sh minui|allium glibc` [Shared]
- **Cache Key:** `ui-glibc-*` (hash of `ui/**`)
- **Submodules:**
  - `minime/ui/minui/` [Shared] (dual musl/glibc)
  - `minime/ui/allium/` [Shared] (dual musl/glibc)
- **GH Artifact:** `ui-glibc`
  - `minui-glibc-aarch64.tar.xz`
  - `allium-glibc-aarch64.tar.xz`

↓ *(Uploads artifacts to GH & Passes to Tier 2)*

3. Artifact Transfer & Caching Layer

- **Actions Used:** `actions/download-artifact@v4` [Shared], `actions/cache/restore@v4` [Shared]
- **Artifacts Downloaded:** `bootloader-out` + matching UI artifact (`ui-musl` for Alpine, `ui-glibc` for Buildroot)
- **Tier 1 Output Caches:** Bootloader Cache (`bootloader-*`), UI Cache (`ui-musl-*`, `ui-glibc-*`)
- **Tier 2 OS Caches:** CCache (`~/alpine-ccache`, `~/.buildroot-ccache`) + DL Cache (`~/alpine-dl`, `~/.buildroot-dl`) [Specific]

↓ *(Fans out to 6 parallel matrix jobs)*

4. Target Component Compilation (build.sh)

4A. Alpine Component Builder (musl)

Alpine (musl) Component Builder (3 Jobs)
- **Command:** `make -C minime/targets/alpine components BOARD=` [Specific]
- **Builder Script:** `minime/targets/alpine/scripts/build.sh` [Specific]
- **Inputs:** `minime/boards//` [Specific], `minime/boards/common/` [Shared]
- **Produced Component Artifacts:**
  - `system.erofs` (Immutable EROFS rootfs with Mesa/Panfrost)
  - `Image` (Linux kernel binary)
  - `initramfs` & `*.dtb` (Device Tree Blobs)

4B. Buildroot Component Builder (glibc)

Buildroot (glibc) Component Builder (3 Jobs)
- **Command:** `make -C minime/targets/buildroot components BOARD=` [Specific]
- **Builder Script:** `minime/targets/buildroot/scripts/build.sh` [Specific]
- **Inputs:** `minime/boards//` [Specific], `minime/boards/common/` [Shared]
- **Produced Component Artifacts:**
  - `system.erofs` (Immutable EROFS rootfs with glibc & Mali drivers)
  - `Image` (Linux kernel binary)
  - `initramfs` & `*.dtb` (Device Tree Blobs)

↓ *(Passes components (system.erofs, Image, DTBs) to Packaging)*

5. Image Assembly & Packaging (mkimage.sh / mkupdate.sh)

5A. Alpine Image Assembly (musl)

Alpine Image & Update Packager (musl)
- **Commands:**
  - `make image BOARD= UI=minui|allium` [Specific]
  - `make update BOARD= UI=minui|allium` [Specific]
- **Packaging Scripts:**
  - `minime/build/genassets.sh` [Shared]
  - `minime/build/mkimage.sh` [Shared]
  - `minime/build/mkupdate.sh` [Shared]
- **Produced Release Archives:**
  - `*.img.xz` (Raw SD card flashable image)
  - `*.tar.xz` (OTA update package)

5B. Buildroot Image Assembly (glibc)

Buildroot Image & Update Packager (glibc)
- **Commands:**
  - `make image BOARD= UI=minui|allium` [Specific]
  - `make update BOARD= UI=minui|allium` [Specific]
- **Packaging Scripts:**
  - `minime/build/genassets.sh` [Shared]
  - `minime/build/mkimage.sh` [Shared]
  - `minime/build/mkupdate.sh` [Shared]
- **Produced Release Archives:**
  - `*.img.xz` (Raw SD card flashable image)
  - `*.tar.xz` (OTA update package)

↓ *(Uploads output archives to GitHub Releases)*

6. Release & Delivery

- **Workflow Step:** `Upload to Testing Release` in `.github/workflows/build.yml` [Shared]
- **Tool / CLI:** `gh release upload testing /*.img.xz /*.tar.xz --clobber` [Shared]
- **Destination:** GitHub Releases — **testing** tag